

AAVEDAN SETU

AI-Powered Civic Grievance Triage & Resolution Platform

Document Type: Website User Interface & Functionality Guide

Target Audience: SIH 2026 Evaluation Panel / Municipal Administrators / System Integrators

Coverage: 12 System Modules, 45+ Buttons, Forms, Coordinates Pickers & AI Conversational Engines

Date of Compilation: August 2026

Status: Production Ready

1. Main Layout: Top Navbar & Navigation Sidebar

The global layout wraps the authenticated user session, offering real-time notifications, multi-lingual translation, and workspace toggles.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
Sidebar Navigation Links	Icon + Text Menu Links	Sidebar	Navigates between Dashboard, Grievance Board, Budgeting portal, Government Schemes, and profile views.
Grievance Status Bell	Bell Icon (Top-Right Navbar)	Header	Toggles an inline panel displaying real-time push notification updates (e.g. ticket updates, resolution comments).
Switch to Officer/Citizen Mode	Floating Gradient Button	Navbar / Dashboard	Demo switch toggling dashboard visualization between Citizen portal and Officer's backend resolution ledger.
Google Translate Dropdown	Dropdown Select	Top Header	Injects Google Translate DOM modifications dynamically translating the entire interface to 12 regional languages.
Profile Menu Badge	User Avatar (Top-Right)	Navbar	Clicking displays a dropdown with options to view Account profile or trigger the Logout action.

2. Landing & Authentication Module

Serves as the gateway for citizen onboarding. Supports password authentication, OTP-based mobile registration, and Google OAuth 2.0.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
Continue with Google	Large Action Button (with G-Logo)	Login / Register	Triggers the Google OAuth accounts popup. Verifies access token on backend, creates DB record and generates SimpleJWT keys.
Password Mode Tab	Pill Segmented Selector	Login / Register	Switches input field bindings to email + password validations.
Login with OTP Tab	Pill Segmented Selector	Login / Register	Switches form bindings to mobile number validation and OTP verification fields.
Send OTP	Action Button (Right inline)	OTP Forms	Validates 10-digit format and calls backend OTP service. Triggers a 60s cooldown timer and displays mock OTP in browser toast for demo.

Verify & Submit	Primary Action Button	OTP Forms	Sends OTP code to verify endpoint. On success, requests SimpleJWT tokens and redirects user.
Forgot Password	Link Label (Below inputs)	Login Page	Routes the user to phone verification layout to send password reset code.
Reset Password / Submit	Primary Action Button	Reset Page	Verifies OTP and new password inputs, calling backend to reset DB password hashes.

3. Citizen Dashboard & Analytics

Provides the home screen layout for authenticated citizens, rendering aggregate charts and quick action portals.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
File New Grievance	Primary Pulsing Button	Main Layout	Opens the dual-triage module to submit complaints.
Budgeting Portal	Outline Action Card	Main Dashboard	Navigates directly to the Civic Budgeting capital allocation board.
Complaint Trend Chart	Interactive Line Chart	Dashboard Centre	Visualizes monthly complaint volumes. Hovering displays specific category data counts.
Department Distribution	Doughnut Chart	Dashboard Right	Breaks down complaints by municipal departments (PWD, Water, Sanitation, Electricity, safety).

4. Grievance Filing Triage — AI Assistant Mode

Aavedan Saathi provides a natural language filing interface using intent categorization and fuzzy location parsing.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
Voice Input (Microphone)	Pulsing Mic Button	Chat Box	Activates browser HTML5 speech-to-text (`webkitSpeechRecognition`) to capture complaints in regional dialects.
Send Message	Arrow Icon Button	Chat Box	POSTs raw input to AI Orchestrator. Checks state variables, extracts location entities, and determines next action.
GPS Auto-Detect Location	Location Pin Icon Button	Chat Box widget	Queries navigator.geolocation API, fetches coordinates, and reverse-geocodes State/District via OpenStreetMap.
Switch to Manual Form	Text Handoff Button	Chat Assistant	Transfers AI-extracted category, district, and address parameters into manual form fields, exiting chat mode.

Preview Official Email	Outline Eye Button	Handoff Screen	Opens a mock-up rendering the structured grievance letter drafted by the AI microservice.
Send Official Email	Gradient Action Button	Handoff Screen	Instructs SMTP handlers to dispatch the grievance report directly to the correct department's division email.

5. Grievance Filing Triage — Manual Form Mode

Enables citizens to bypass conversational filing and manually register grievances with precise geographic coordinates and attachments.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
State Dropdown	Dropdown Selection	Manual Form	Populates all 36 Indian States/UTs. Selecting filters District selection.
District Dropdown	Dropdown Selection	Manual Form	Populates official districts based on State (cascading). Typo-tolerant matching checks inputs on submission.
Map Coordinate Picker	Interactive Leaflet Container	Manual Form	Displays map centered on user GPS. Clicking drops a pin, auto-filling precise Latitude & Longitude decimal values.
Evidence Photo Upload	Multi-File Input Area	Manual Form	Supports drag-and-drop file uploads. Processes image compression before storing on backend.
Submit Complaint	Primary Action Button	Form Footer	Validates inputs, maps coordinates, and inserts ticket record in database with generated reference number (e.g. GC-2026-000001).

6. My Complaints & Grievance Board

The central repository displaying active, pending, and resolved complaints with multi-dimensional filtering.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
Filter Board Accordeon	Toggle Panel Button	Board Header	Expands filter criteria including Status, Priority, Category, Department, State, and District.
Toggle Upvote / Support	Heart / Up-Arrow Button	Complaint Cards	Adds citizen vote weight to complaint. Toggles state and sends HTTP request to DB, triggering live project score recalculations.
View Detail Board	Arrow Icon Button	Complaint Cards	Routes browser to full detail page of selected grievance ticket.

7. Complaint Detail & Resolution Ledger

Verifies resolution progress. Integrates the Officer Resolution Panel and the Citizen Verification Ledger.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
Resolve Grievance (Officer)	Action Button	Officer Panel	Opens upload dialog for department officers to select 'After Repair' proof images.
Submit Group Resolution	Action Button	Officer Panel	Saves resolution proof. Cascades status to ALL complaints grouped within the ward cluster, marking them Under Review.
Verify Work Done (Citizen)	Success Check Button	Citizen Ledger	Citizen approves proof. Accumulating 3 approvals updates CivicProject status to COMPLETED and resolves all complaints.
Reject Proof (Citizen)	Rejection Cross Button	Citizen Ledger	Citizen rejects proof, prompting reason dialog. 3 rejections reopens the tickets to IN_PROGRESS and alerts the resolving officer.

8. Civic Budgeting & Community Projects Portal

Empowers citizens to democratically allocate municipal capital from the ₹5 Crore District Repair Pool.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
Vote to Fund Project	Large Check Box Toggle	Budgeting Cards	Toggles user vote on a CivicProject. Triggers atomic database transaction syncing priority scores and percentages.
District Filter	Select Dropdown	Budgeting Header	Dynamically updates unallocated backlog cost counters based on selected district database query.

9. Government Schemes & Profile Modules

AI schemes matching engine and citizen profile panel.

Button / Element Name	UI Placement / Trigger	Interactive Context	Internal / System Functionality
Check Eligibility	Action Button	Schemes Page	Evaluates user age, income, category, and state against welfare rules. Filters ineligible state yojanas.

Update Profile Detail	Primary Action Button	Profile Page	Saves user profile modifications, updating email, phone, and full name values in backend DB User model.
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